

Week 5 AI-Assisted Emergency Department Triage Dataset - Feasibility Memo

Prepared for: Dr. De Fretias & Emergency Department Board

Executive Verdict

The de-identified emergency department dataset is suitable for developing a baseline AI-assisted triage prediction model, provided that data preprocessing, feature engineering, and clinical validation are completed before model development.

Dataset Summary

This exploratory analysis assessed a real-world emergency department dataset containing 55,121 patient encounters and 226 clinical variables. The dataset includes demographic information, arrival characteristics, triage vital signs, clinical disposition, and chief complaint indicators. The target variable is the Emergency Severity Index (ESI).

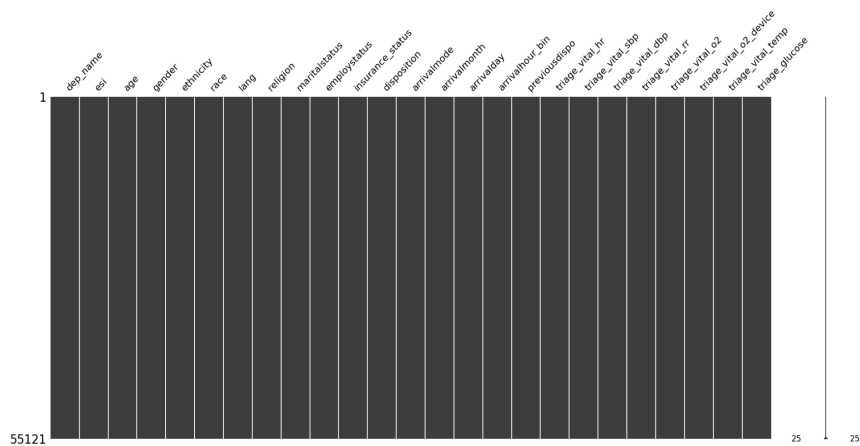
Most patients fall within ESI Levels 2–4, while comparatively fewer belong to Levels 1 and 5, indicating class imbalance that should be considered during model development. Overall, the dataset provides a clinically rich foundation for predictive modelling.

Data Quality Assessment

Missing Data

Core structured variables demonstrated high completeness with minimal systematic missingness, supporting the dataset's suitability for predictive modelling.

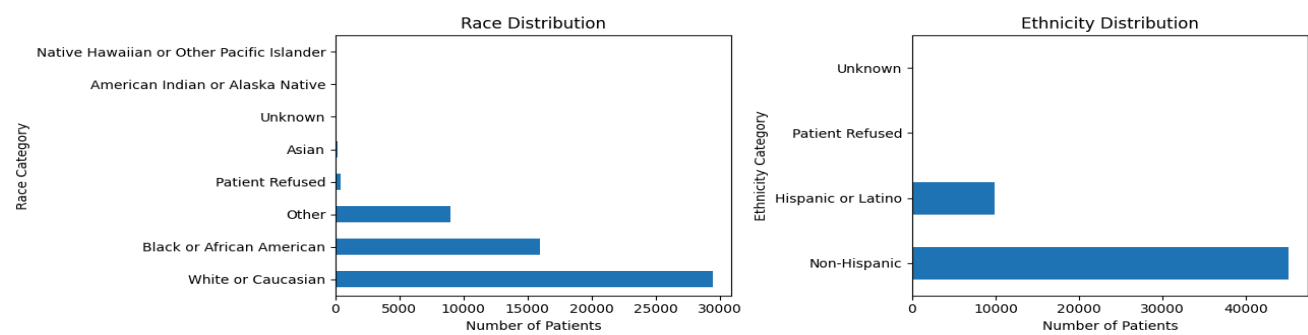
Figure 1. Missingness Matrix



Demographic Characteristics

Demographic variables, including age, gender, race, and ethnicity, were explored to assess population characteristics and data completeness. These variables are important for understanding the patient population and evaluating model fairness.

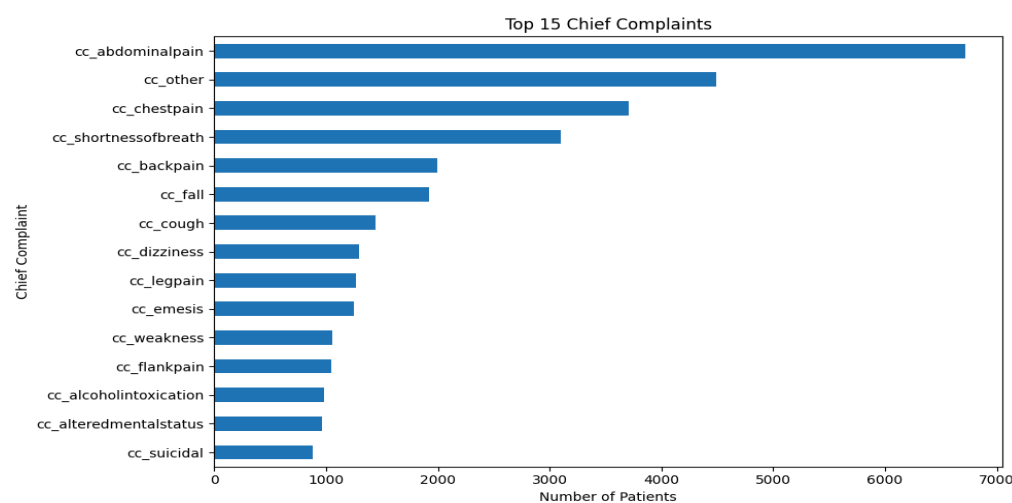
Figure 2. Race and Ethnicity Distribution



Chief Complaint Distribution

Chief complaint indicators demonstrated a typical long-tail distribution, with several complaints occurring much more frequently than others. These variables are expected to provide valuable clinical context during triage prediction.

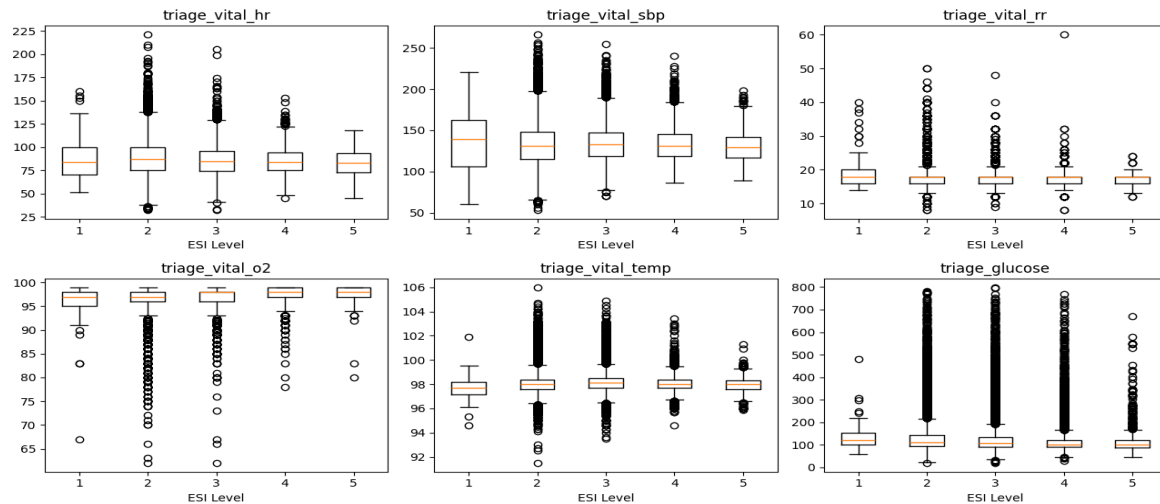
Figure 3. Top Chief Complaints



Vital Signs

Triage vital signs varied across ESI categories and demonstrated clinically plausible patterns. Several outliers were observed and should be validated before modelling.

Figure 4. Distribution of Triage Vital Signs Across ESI Levels



Boxplots illustrate the distributions of heart rate, systolic blood pressure, respiratory rate, oxygen saturation, temperature, and glucose across ESI categories. Differences in these physiological measurements support their potential usefulness as predictors in future triage modelling while also highlighting clinically plausible outliers that should be reviewed during data preprocessing.

Correlation Analysis: A preliminary correlation assessment of selected triage vital signs and the ESI outcome indicated generally weak to moderate linear relationships. This suggests that triage decisions are influenced by multiple interacting clinical variables rather than a single physiological measurement. Consequently, multivariable modelling will be more appropriate than relying on individual predictors.

Top 3 Reasons to Proceed

Reason	Summary
Rich clinical information	Demographics, vital signs and chief complaints provide clinically relevant predictors.
High data completeness	Structured variables contain minimal missingness.
Large sample size	Over 55,000 encounters support robust model development.

Top 3 Data Quality Concerns

Concern	Impact
Class imbalance	May reduce prediction performance for rare ESI categories.
High-dimensional complaints	Requires feature engineering.
Clinical outliers	Requires validation before modelling.

Top 10 Candidate Predictive Features

Rank	Feature	Clinical Justification
1	Respiratory Rate	Strong indicator of respiratory compromise and acute physiological deterioration.
2	Heart Rate	Reflects haemodynamic stress and illness severity.
3	Oxygen Saturation	Critical measure of respiratory function and oxygenation.
4	Systolic Blood Pressure	Hypotension is associated with severe illness requiring urgent care.
5	Temperature	Supports identification of infection and systemic inflammatory responses.
6	Age	Older patients frequently require higher-acuity assessment because of increased clinical risk.
7	Chief Complaint	Provides immediate clinical context guiding triage decisions.
8	Pain Score	Influences urgency assessment and resource allocation.
9	Arrival Mode	Ambulance arrivals often indicate greater illness severity.
10	Previous Disposition	May provide additional context regarding recent healthcare utilisation.

Limitations

Several limitations should be acknowledged before proceeding to predictive modelling.

First, this dataset represents a single healthcare institution, limiting external generalisability. Second, although demographic completeness is excellent, high-dimensional chief complaint variables require additional preprocessing. Finally, model fairness should be assessed carefully to ensure that demographic variables do not contribute to unintended bias.

Recommendation to the ED Board

Based on the exploratory analysis, this dataset is **suitable for baseline AI-assisted triage modelling**.

The combination of a large sample size (55,121 encounters), excellent completeness of structured variables, clinically meaningful vital signs, and comprehensive chief complaint information provides a strong foundation for predictive modelling.

Before implementation, the following actions are recommended:

1. Validate clinically implausible observations.
2. Address class imbalance during model development.
3. Engineer and reduce high-dimensional chief complaint variables.
4. Evaluate fairness across demographic subgroups.
5. Perform external validation using data from additional healthcare institutions before clinical deployment.